

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 6 : REACTIVITY IN THE EXACT PKEs

EFFECTIVE DELAYED NEUTRON FRACTIONS

POINT REACTOR MODEL

THE PROMPT NEUTRON BALANCE EQUATION

$$\frac{1}{\bar{v}\nu\Sigma_f} \frac{d\hat{\Phi}}{dt} = \rho\hat{\Phi}$$

$$\frac{d\hat{\Phi}}{dt} = \frac{\rho}{\Lambda} \hat{\Phi}$$

$$\frac{d\hat{\Phi}}{dt} = \frac{\Delta k}{\ell} \hat{\Phi}$$

THE INTUITIVE POINT KINETICS EQUATIONS

$$\frac{d\hat{\Phi}}{dt} = \frac{\rho - \beta}{\Lambda} \hat{\Phi} + \frac{1}{\Lambda\nu\Sigma_f} \sum_k \lambda_k \hat{C}_k$$

$$\frac{d\hat{C}_k}{dt} = -\lambda_k \hat{C}_k + \nu_{dk}\Sigma_f \hat{\Phi}, \quad k = 1, \dots, 6$$

ONE-GROUP POINT KINETICS EQUATIONS

$$\Lambda \frac{dp}{dt} = (\rho - \beta)p + \frac{1}{\hat{S}_{f0}} \sum_k \lambda_k \hat{C}_k + \frac{\hat{S}}{\hat{S}_{f0}}$$

$$\Lambda = \frac{1}{\nu \Sigma_f} \left(\frac{\bar{1}}{\bar{v}} \right)$$

$$\hat{S}_{f0} = \nu \Sigma_f \hat{\Phi}_0$$

APPROXIMATE POINT KINETICS EQUATIONS

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \frac{1}{\Lambda} \sum_k \lambda_k \zeta_k + \frac{1}{\Lambda} s(t)$$

$$\frac{d\zeta_k}{dt} = -\lambda_k \zeta_k + \beta_k p(t)$$

POINT KINETICS EQUATIONS

The more familiar form of the [point kinetics equations](#) is:

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \sum_k \lambda_k c_k + s_c$$
$$\frac{dc_k}{dt} = -\lambda_k c_k + \frac{\beta_k}{\Lambda} p$$

where:

$$c_k(t) = \frac{1}{\Lambda} \zeta_k(t) \quad s_c(t) = \frac{1}{\Lambda} s(t)$$

EXACT POINT KINETICS EQUATIONS

$$\phi(\mathbf{r}, E, t) = p^w(t) \cdot \Psi^w(\mathbf{r}, E, t)$$

$$\begin{aligned} \int_V \int_0^\infty \frac{w(\mathbf{r}, E)}{\nu(E)} \frac{\partial \phi(\mathbf{r}, E, t)}{\partial t} dE dV &= \frac{dp(t)}{dt} \int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV \\ &+ p(t) \frac{d}{dt} \int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV \end{aligned}$$

$$\int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV = \text{constant}$$

Flux amplitude function:

$$\dot{p}(t) = \frac{\rho(t) - \beta(t)}{\Lambda(t)} p(t) + \frac{1}{\Lambda_0} \sum_k \lambda_k \zeta_k(t)$$

Reduced precursors:

$$\dot{\zeta}_k(t) = -\lambda_k \zeta_k(t) + \frac{F(t)}{F_0} \beta_k(t) p(t)$$

$$\beta_k(t) = \frac{1}{F(t)} (\Phi_0^*, \mathbf{F}_{dk} \Psi)$$

$$\zeta_k(t) = \frac{1}{F_0} (\Phi_0^*, \chi_{dk} C_k)$$

Reactivity in the Exact PKEs

DYNAMIC AND STATIC REACTIVITY

Is the reactivity of 1\$ a fixed number?

- No: by definition a reactivity of 1\$ is equal to the delayed neutron fraction, β ; and the value of β depends on the approximations applied in calculating it.
- The exact value of β is given by $\beta(t) = \frac{1}{F(t)} (\Phi_0^*, \mathbf{F}_d \Psi) = \sum_k \beta_k(t)$
- An often applied approximation is $\beta(t) = \beta_0$

The static reactivity is given by: $\rho^{st} = \frac{(\Phi_0^*, [\Delta \mathbf{F} - \Delta \mathbf{M} \Phi])}{(\Phi_0^*, \Delta \mathbf{F} \Phi)}$, where “0” represents a specific critical state

The reactivity in the exact kinetics equation, $\rho(t) = \frac{1}{F(t)} (\Phi_0^*, [\Delta \mathbf{F} - \Delta \mathbf{M}] \Psi)$, is called “dynamic reactivity”:

The dynamic reactivity is formed in fully analogy to the exact static reactivity with the difference of:

ρ^{dyn} is formed with the time-dependent flux $\phi(\mathbf{r}, E, t)$ as it physically appears during a transient;
 ρ^{st} is formed with the λ mode, $\Phi_\lambda(\mathbf{r}, E)$, of the “perturbed” system

REACTIVITY IN THE EXACT PKE

If the flux and adjoint flux used in the exact point kinetics equations are inaccurate, the resultant reactivity will also be inaccurate.

- This inaccuracy comes from the zero-order terms $(\Phi_0^*, \mathbf{F}_0, \Psi)$ and $(\Phi_0^*, \mathbf{M}_0, \Psi)$:
 - *For a critical reactor $(\Phi_0^*, \mathbf{F}_0, \Psi) = (\Phi_0^*, \mathbf{M}_0, \Psi)$*
 - *If Φ_0^* is not accurate, there will be an erroneous finite residual leading to an error in the reactivity*
- The classical conceptual idea for eliminating the error caused by zero-order terms is to cancel analytically the zero-order terms in the perturbation formula:
 - *By doing so, the first-order perturbation (i.e., ρ) is determined by the first-order operator changes $\Delta \mathbf{F}$ and $\Delta \mathbf{M}$ rather than as the difference between large zero-order terms*

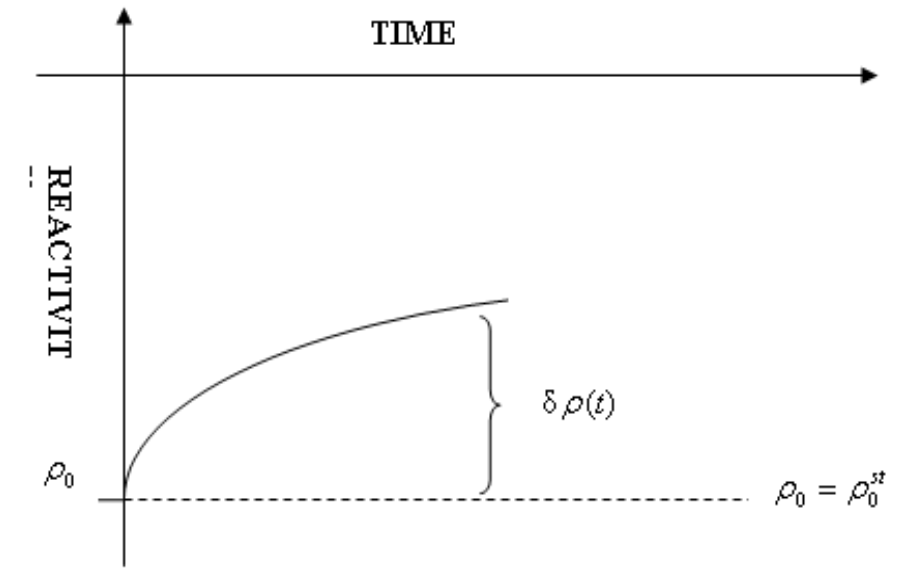
REACTIVITY IN THE EXACT PKE

The reactivity can be expressed as a sum of two parts:

$$\rho(t) = \rho_0 + \delta\rho(t),$$

where $\delta\rho(t)$ describes the reactivity increment about ρ_0 .

The initial reactivity ρ is identical with the static reactivity, $\rho_0 = \rho_0^{st}$.



Description of the reactivity $\rho(t)$ in subcritical reactors

The reactivity increment is given by:

$$\delta\rho(t) = \frac{1}{F_\lambda(t)} (\Phi_{\lambda_0}^*, [\lambda_0 \mathbf{F} - \mathbf{M}] \Psi)$$

REACTIVITY IN THE EXACT PKE

Note that:

- When applying an energy-dependent weighting function, $\Phi_0^*(\mathbf{r}, E)$, the prompt and delayed neutron emission spectra are not eliminated by the energy integration.
- Also, the difference between the importance of neutrons before and after scattering has an effect on the reactivity.
- The independent source is also weighted by the importance of a neutron injected into the reactor at various locations and with various energies.

Effective Delayed Neutron Fractions

THE EFFECTIVE β VS. ONE-GROUP β

$$\beta_k(t) = \frac{1}{F(t)} (\Phi_0^*, \mathbf{F}_{dk} \Psi)$$

The “effective” delayed neutron fraction, $\beta(t)$, differs numerically from the corresponding quantity without adjoint neutron weighting, i.e., from the one-group β :

1. The energy dependence of the adjoint flux provides the proper importance weighting for the *emission spectra of the delayed neutrons*;
2. The space dependence “weighs” the *spatial distribution of the delayed neutron emission*.

THE EFFECTIVE β VS. ONE-GROUP β

Let's apply a separation approximation for the initial adjoint flux: $\phi_0^*(\mathbf{r}, E) = \phi_0^*(\mathbf{r})\varphi^*(E)$

First, consider the simple case of a [single fissionable isotope](#) and assume time-independent ψ :

$$\beta_k(t) = \frac{(\Phi_0^*, \mathbf{F}_{dk} \Psi)}{(\Phi_0^*, \mathbf{F}_k \Psi)}$$

$$\beta_k = \frac{\int_E \chi_{dk}(E) \phi_0^*(E) dE \int_V \phi_0^*(\mathbf{r}) \nu_{dk} \int_{E'} \Sigma_f(\mathbf{r}, E') \psi(\mathbf{r}, E') dE' dV}{\int_E \chi(E) \phi_0^*(E) dE \int_V \phi_0^*(\mathbf{r}) \bar{\nu} \int_{E'} \Sigma_f(\mathbf{r}, E') \psi(\mathbf{r}, E') dE' dV}$$

$\bar{\nu}$ and ν_{dk} are only weakly dependent on space $\rightarrow$ if this weak space dependence is neglected, the space and E' integrals in the numerator and denominator cancel and one obtains:

$$\beta_k \approx \bar{\beta}_k \gamma_{dk}, \text{ where } \bar{\beta}_k = \frac{\nu_{dk}}{\bar{\nu}} \text{ and } \gamma_{dk} = \frac{\int \chi_{dk}(E) \phi_0^*(E) dE}{\int \chi(E) \phi_0^*(E) dE}$$

For a single fissionable isotope, the “effective” and one-group β 's differ by the ratio of the importance weight of the delayed and total fission neutron emission spectra, γ_{dk} .

EFFECTIVE DELAYED NEUTRON FRACTIONS

The following table gives the values of the ratios of the importance weights of delayed and total neutron emission spectra for a typical large fast breeder reactor.

The average importance of the delayed neutrons is $\sim 18\%$ smaller than that of the total number of fission neutrons.

Delayed Neutron Group, k	Ratio of the Relative Importance of Delayed and Prompt Fission Neutrons, γ_{dk}
1	0.802
2	0.831
3	0.818
4	0.825
5	0.825
6	0.825

EFFECTIVE DELAYED NEUTRON FRACTIONS

The reason for the small variation in the importance of the different delay groups and a large difference between delayed and prompt neutron importance is that delayed neutrons in all delay groups have only a small chance of causing fission in ^{238}U since most delayed neutrons have energies below the threshold in the ^{238}U fission cross-section.

This is reflected in a lower importance of $\chi_d(E)$ compared to $\chi(E)$.

In addition, $\eta(E) = \frac{\nu(E)\sigma_f(E)}{\sigma_a(E)}$ of ^{239}Pu increases with energy, which also gives χ a higher importance than χ_d .

These effects cause delayed neutrons to have an $\sim 18\%$ smaller importance than prompt neutrons in fast reactors.

EFFECTIVE DELAYED NEUTRON FRACTIONS

In a thermal reactor, the adjoint spectrum generally increases with decreasing energy because of increasing proximity to the range of the thermal fission.

Fast fission is of minimal importance due to the rapid slowing down of the neutrons below the fast fission threshold in ^{238}U .

Thus, the increase of $E\varphi_0^*(E)$ with decreasing E prevails in the determination of the average importance of delayed and prompt fission neutrons.

Therefore, delayed neutrons have a larger importance than prompt neutrons in a thermal reactor.

EFFECTIVE DELAYED NEUTRON FRACTIONS

For a reactor with two fissionable isotopes, such as ^{239}Pu and ^{238}U :

$$\beta_k \cong \frac{\int_E \chi_{dk}(E) \phi_0^*(E) dE}{\int_E \chi(E) \phi_0^*(E) dE} \times \frac{\int_V \phi_0^*(\mathbf{r}) [\nu_{dk9} R_{f9}(\mathbf{r}) + \nu_{dk8} R_{f8}(\mathbf{r})] dV}{\int_V \phi_0^*(\mathbf{r}) \bar{\nu}(\mathbf{r}) R_f(\mathbf{r}) dV}$$

where R represents an energy-integrated reaction rate.

EFFECTIVE DELAYED NEUTRON FRACTIONS

$$\beta_k = \frac{\int_E \chi_{dk}(E) \phi_0^*(E) dE \int_V \phi_0^*(\mathbf{r}) \nu_{dk} \int_{E'} \Sigma_f(\mathbf{r}, E') \psi(\mathbf{r}, E') dE' dV}{\int_E \chi(E) \phi_0^*(E) dE \int_V \phi_0^*(\mathbf{r}) \bar{\nu} \int_{E'} \Sigma_f(\mathbf{r}, E') \psi(\mathbf{r}, E') dE' dV}$$

$$\beta_k \cong \frac{\int_E \chi_{dk}(E) \phi_0^*(E) dE}{\int_E \chi(E) \phi_0^*(E) dE} \times \frac{\int_V \phi_0^*(\mathbf{r}) [\nu_{dk9} R_{f9}(\mathbf{r}) + \nu_{dk8} R_{f8}(\mathbf{r})] dV}{\int_V \phi_0^*(\mathbf{r}) \bar{\nu}(\mathbf{r}) R_f(\mathbf{r}) dV}$$

Comparing the β_k equations for the single- and two-isotopes cases, one observes that:

1. Both denominators are essentially the same except that in the two-isotopes case both isotopes contribute to $\bar{\nu}(\mathbf{r}) R_f$;
2. In the numerator, the ^{238}U effect is strongly amplified by its high delayed neutron yield compared to ^{239}Pu ;
3. Removing $R_{f9}(\mathbf{r})$ from the bracket in two-isotopes equation yields an expression that shows the strong space, delay group, and system dependence of the contribution of ^{238}U to the delayed neutron production:

$$\tilde{\nu}_{dk}(\mathbf{r}) = \nu_{dk9} + \nu_{dk8} \frac{R_{f8}(\mathbf{r})}{R_{f9}(\mathbf{r})}$$

EFFECTIVE DELAYED NEUTRON FRACTIONS

Comparison of ²³⁹Pu and ²³⁸U Contributions to the Production Rate of Delayed Neutrons

	Core Center	Core/Blanket Interface
	Percentage Addition of ²³⁸ U-Produced Delayed Neutrons to ²³⁹ Pu Contribution	
Delayed Neutron Group, <i>k</i>	$100 \frac{\nu_{dk8} R_{f8}}{\nu_{dk9} R_{f9}}$	
1	29	17
2	44	26
3	71	42
4	105	62
5	182	108
6	315	185
	$\frac{R_{f8}}{R_{f9}}$	
	0.142	0.0836

The relative ²³⁸U contribution strongly increases with the decreasing lifetime of the precursor groups from 29 to 315 % in the core center.

This is explained by the dependence of ν_{dk} for the ²³⁸U and ²³⁹Pu.

The bottom line of the table gives the ratios of the fission rates in ²³⁸U and ²³⁹Pu in a FBR core center and at the core/blanket interface, where the neutron spectrum is much softer than in the center.

The reduction of the relative fission rate across the fast reactor core amounts to more than a factor of 1.5.

This reduction is reflected also in the relative delayed neutron contributions as shown in columns 2 and 3.

EFFECTIVE DELAYED NEUTRON FRACTIONS

The adjoint flux weighting in the two-isotopes equation amplifies the increase of $\tilde{v}_{dk}$ values toward the core center, since the importance of neutrons increases toward the center of the core.

Thus, spatial weighting with the adjoint function increases the effective delayed neutron fraction in fast reactors (since it provides a higher weight for the higher ^{238}U contributions near the center of the core).

The total reduction of β , resulting from the energy dependence of $\Phi^*(\mathbf{r}, E)$, is generally larger than the increase caused by the spatial weighting.

For long-lived delayed neutrons groups, the increase due to spatial adjoint flux weighting could be larger than the decrease due to spectral adjoint flux weighting.

EFFECTIVE DELAYED NEUTRON FRACTIONS

The two-isotopes equation for β_k can be written in terms of one-group β 's as:

$$\beta_k = \gamma_{dk} \left(\beta_{k9} \frac{S_{f9}^*}{S_f^*} + \beta_{k8} \frac{S_{f8}^*}{S_f^*} \right) \quad \text{with} \quad S_{fi}^* = \int \phi_0^*(\mathbf{r}) R_{fi}(\mathbf{r}) dV$$

The expression in parentheses describes the effect of spatial adjoint flux weighting whereas γ_{dk} contains the effect of spectral weighting.

Point Reactor Model

THE POINT REACTOR MODEL

The integral kinetics parameters ρ , β_k and Λ are defined as integrals over space and energy of integrands that depend on the time-dependent space function $\psi(\mathbf{r}, E, t)$:

If $\psi(\mathbf{r}, E, t)$ could be calculated and used to find $\rho(t)$, $\beta_k(t)$ and $\Lambda(t)$ from their “exact” definitions, they can be used in the exact point kinetics equations to determine the amplitude function $p(t)$.

Generating such shape functions would require solutions of full $(\mathbf{r}, E, t)$ - dependent problems.

Approximate methods for such calculations comprise what is called “space-energy dependent dynamics”.

THE POINT REACTOR MODEL

The most commonly used approximate method is called the “[point reactor model](#)”:

The kinetics equations for the point reactor model are normally referred to as “point kinetics equations”;

They can be obtained from the exact point kinetics equations by introducing [one major](#) and [two minor](#) simplifications.

[The major simplification](#) consists of neglecting the time dependence of the flux shape; that is the initial flux shape is used in forming the integral kinetics - $\psi(\mathbf{r}, E, t) \cong \phi_0(\mathbf{r}, E)$:

If $p_0 = 1$, the initial flux shape is equal to the initial flux $\Phi_0(\mathbf{r}, E)$.

The time dependence of the neutron flux is thus assumed separable from the $(\mathbf{r}, E)$ dependence.

THE POINT REACTOR MODEL

There two minor simplifications are:

1. The arbitrary denominator $F(t)$ is replaced by its initial value:

$$F(t) \rightarrow F_0 = (\Phi_0^*, \mathbf{F}_0 \Psi_0)$$

2. The delayed fission neutron operator $\mathbf{F}_d$ is approximated by the initial operator:

$$\mathbf{F}_d \cong \mathbf{F}_{d0}$$

THE POINT REACTOR MODEL

These three simplifications have the following consequences:

1. If the reactor is initially critical, a substitution of the last two equations into the time-dependent exact reactivity formula converts it into the “first-order perturbation theory formula”:

$$\rho(t) = \rho^{(1)}(t) = \frac{1}{F_0} (\Phi_0^*, [\Delta \mathbf{F} - \Delta \mathbf{M}] \Phi_0).$$

2. Substitution of $F(t)$ by F_0 into the time-dependent exact generation time formula yields:

$$\Lambda(t) \rightarrow \Lambda_0 = \frac{K_0}{F_0} \qquad \frac{F_0}{F(t)} \rightarrow 1$$

3. The approximation of the delayed fission neutron operator by its initial operator leads to the neglect of the (weak) time dependence of the delayed neutron fractions:

$$\beta_k(t) \rightarrow \beta_{k0} = \frac{1}{F_0} (\Phi_0^*, \mathbf{F}_{dk0} \Phi_0)$$

THE POINT REACTOR MODEL

The implementation of these approximations and the replacements into the exact point kinetics equations yields what is usually referred to as the [point kinetics equations](#):

$$\dot{p}(t) = \frac{\rho(t) - \beta}{\Lambda} p(t) + \frac{1}{\Lambda} \sum_k \lambda_k \zeta_k(t)$$

$$\dot{\zeta}_k(t) = -\lambda_k \zeta_k(t) + \beta_k p(t)$$

Note that this form of the exact point kinetics equation contains the initial values of ρ , β_k , and Λ (this is not indicated by an index 0 in order to avoid indices).

THE POINT REACTOR MODEL

The neglect of the time dependence of the flux shape in the point reactor model is usually understood to be an assumption for the “gross neutron flux”.

The “fine structure” of the flux, which is often accounted for by “effective cross sections”, is allowed to depend on time.

The most important “fine structure” of the flux, which must be considered time-dependent, is the spectrum deformation caused by the Doppler broadening of resonances.

This resonance broadening is a function of the temperature, and the temperature depends on time during a transient.

The time-dependent spectral deformation due to Doppler broadening is combined with the resonance cross sections and averaged over group intervals.

The results are “effective” group cross sections, which depend on temperature and thus on time:

$$\sigma_g[T(t)]\phi_g(\mathbf{r}, t = 0).$$

Reaction rates are formed by multiplication of the effective, time-dependent, group cross sections with the time-independent gross flux.

THE POINT REACTOR MODEL

- Generally, point kinetics is a good approximation when the flux shape is largely invariable during a transient.
- This is the case for tightly coupled small reactors or for transients in large reactors if they involve fairly uniform perturbations.
- Transients in large reactors (for example, 1000 MW(e) LWR, in which distant areas of the core are only loosely coupled to each other) that are caused by non-uniform perturbations generally require treatment by a space-energy-dependent kinetics method.

Next Class *Prompt Reactivity Feedback Phenomena: Core Power Models and Fuel Temperature Calculations*